

Exam Date & Time: 16-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY BENGALURU
SEVENTH SEMESTER B.TECH DGREE EXAMINATIONS NOVEMBER 2025

Social Networks [CSE_4204]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate how variations in the **structure and type of connections** within a network influence its overall **behavior, stability, and connectivity patterns**. Assess the impact of factors such as **density, clustering, and degree distribution** on information flow and resilience, and **propose** strategies to enhance network efficiency or robustness based on the principles of **network science**. (5)
 - A)
 - B) Apply the process of network data collection from real-world sources such as social media or biological databases, and interpret how the quality and format of the collected data influence the reliability of subsequent network modeling. (3)
 - C) Apply your understanding of **network analysis tools** such as **SNAP** and **Gephi** to **illustrate** how software-based visualization enhances the **comprehension of large-scale network structures**. (2)
- 2) Evaluate how the **balance between strong and weak ties** affects the **stability, cohesion, and adaptability** of a social system. Assess potential risks of over-reliance on either relationship type and **propose** strategies for maintaining an optimal mix of ties to promote robust social connectivity. (5)
 - A)
 - B) Identify an approach to **measure homophily** in a real-world social network using attributes such as **age, profession, or location**, and **interpret** what high or low homophily values indicate about the **diversity** of the network. (3)
 - C) Construct a simple triadic relationship among three individuals to **illustrate** how balance or conflict can emerge within a social structure. (2)
- 3) Compare and **contrast** the results obtained from **degree** and **eigenvector centrality** when analyzing a social or communication network. Assess how each measure captures a different aspect of **node importance and network hierarchy**. (5)
 - A)
 - B) Apply the **PageRank algorithm** to **develop** a simple example network and **interpret** how node ranking changes with variations in link structure, demonstrating its role in **information retrieval** and **web search optimization**. (3)
 - C) Construct a simple example to **illustrate** how the **threshold model** determines whether a user adopts a new idea based on the number of influenced neighbors. (2)

- 4) Evaluate the **significance of power-law distributions** in understanding **real-world networks** such as the web, citation networks, or social media. **Assess** how the presence of a few highly connected nodes influences **network stability, vulnerability, and information flow**. (5)
- A)
- B) Apply the **SI, SIS, and SIR epidemic models** to **illustrate** how disease or information spreads over time within a network, and **interpret** how recovery and reinfection rates influence the overall diffusion process. (3)
- C) Apply the concept of a **scale-free network to identify** how a few highly connected nodes (hubs) influence the **structure and connectivity** of the network. (2)
- 5) Compare the **Watts-Strogatz small-world model** with a **random network model** to **evaluate** how features such as **short average path lengths** and **high clustering coefficients** explain the **balance between randomness and structure** in real networks. (5)
- A)
- B) Develop an example of **viral content propagation** in an online social platform to **interpret** how network structure and **peer influence** contribute to the **speed and reach** of virality. (3)
- C) Illustrate, with a brief example, how **digital marketing campaigns** make use of **network** influence to enhance the **virality** of promotional content. (2)

-----End-----

Event Name		Subject	
058-END SEMSTER-ODD-2025		CSE_4204:Social Networks	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name		Subject	
058-END SEMSTER-ODD-2025		CSE_4204:Social Networks	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)

3	Application	10	25(50%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)